

Elapsed Time Service

Bluetooth® Service Specification

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Abstract:

This specification defines a time service suitable for simple embedded devices that keep track of time using a tick counter or that are aligned with Coordinated Universal Time (UTC) and that maintain an offset for time zone (TZ) or Daylight Saving Time (DST) adjustment. The time stamp format uses a 3-byte value for making calculations easy and provides sufficient metadata for a client to obtain a good understanding of the device clock status.



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1 Introduction

The Elapsed Time Service (ETS) specification defines a GATT service that can be used by personal health devices and other devices that need to keep track of time. The purpose of ETS is to be usable beyond the healthcare domain and to provide:

- A simple time format of a device clock by using a time value that is a counter of time units.
- Time stamps that are understandable stand-alone by having sufficient metadata.

ETS includes one characteristic that reports the current time, status, and capability of the server's clock. The Current Elapsed Time characteristic supports reading, indicating, and writing (optionally when the server allows a client to set the clock). The client can use a write action to set the current time of the sensor.

The time format can be used for time stamps and comes with sufficient metadata on the status of the clock to be fully understood by clients without needing further information from the server.

1.1 Language

1.1.1 Language conventions

In the development of a specification, the Bluetooth SIG has established the following conventions for use of the terms “*shall*”, “*shall not*”, “*should*”, “*should not*”, “*may*”, “*must*”, and “*can*”. In this Bluetooth specification, the terms in [Table 1.1](#) have the specific meanings given in that table, irrespective of other meanings that exist.

Term	Definition
shall	—used to express what is required by the specification and is to be implemented exactly as written without deviation
shall not	—used to express what is forbidden by the specification
should	—used to express what is recommended by the specification without forbidding anything
should not	—used to indicate that something is discouraged but not forbidden by the specification
may	—used to indicate something that is permissible within the limits of the specification
must	—used to indicate either: 1. an indisputable statement of fact that is always true regardless of the circumstances 2. an implication or natural consequence if a separately-stated requirement is followed
can	—used to express a statement of possibility or capability

Table 1.1: Language conventions terms and definitions

1.1.1.1 Implementation alternatives

When specification content indicates that there are multiple alternatives to satisfy specification requirements, if one alternative is explained or illustrated in an example it is not intended to limit other alternatives that the specification requirements permit.



1.1.1.2 Discrepancies

It is the goal of Bluetooth SIG that specifications are clear, unambiguous, and do not contain discrepancies. However, members can report any perceived discrepancy by filing an erratum and can request a test case waiver as appropriate.

1.1.2 Reserved for Future Use

Where a field in a packet, Protocol Data Unit (PDU), or other data structure is described as "Reserved for Future Use" (irrespective of whether in uppercase or lowercase), the device creating the structure shall set its value to zero unless otherwise specified. Any device receiving or interpreting the structure shall ignore that field; in particular, it shall not reject the structure because of the value of the field.

Where a field, parameter, or other variable object can take a range of values, and some values are described as "Reserved for Future Use," a device sending the object shall not set the object to those values. A device receiving an object with such a value should reject it, and any data structure containing it, as being erroneous; however, this does not apply in a context where the object is described as being ignored or it is specified to ignore unrecognized values.

When a field value is a bit field, unassigned bits can be marked as Reserved for Future Use and shall be set to 0. Implementations that receive a message that contains a Reserved for Future Use bit that is set to 1 shall process the message as if that bit was set to 0, except where specified otherwise.

The acronym RFU is equivalent to Reserved for Future Use.

1.1.3 Prohibited

When a field value is an enumeration, unassigned values can be marked as "Prohibited." These values shall never be used by an implementation, and any message received that includes a Prohibited value shall be ignored and shall not be processed and shall not be responded to.

Where a field, parameter, or other variable object can take a range of values, and some values are described as "Prohibited," devices shall not set the object to any of those Prohibited values. A device receiving an object with such a value should reject it, and any data structure containing it, as being erroneous.

"Prohibited" is never abbreviated.

1.2 Table requirements

Requirements in tables in this document are defined as "Mandatory" (M), "Optional" (O), "Excluded" (X), "Not Applicable" (N/A), or "Conditional" (C.n). Conditional statements (C.n) are listed directly below the table in which they appear.

1.3 Conformance

Each capability of this specification shall be supported in the specified manner. This specification may provide options for design flexibility, because, for example, some products do not implement every portion of the specification. For each implementation option that is supported, it shall be supported as specified.

1.4 Byte transmission order

All characteristics used with this service shall be transmitted with the least significant octet (LSO) first (i.e., little endian). Where the format is described in tables in this document, the LSO is the first octet in the topmost field of the table; the most significant octet (MSO) is the last octet in the bottommost field of

the table. Where characteristics are defined in the GATT Specification Supplement (GSS), see Section 2.2 in GSS [\[3\]](#).



2 Service

2.1 Service dependencies

ETS does not depend on other services.

2.2 Bluetooth Core Specification release compatibility

This specification is compatible with Bluetooth Core Specification v4.2 or later [1] that includes the Generic Attribute Profile (GATT).

2.3 Transport dependencies

There are no transport-related dependencies.

2.4 Attribute Protocol Application error codes

This service references Common Profile and Service Error Codes that are defined in Part B, Section 1 of the Core Specification Supplement (CSS) [2]. This service also defines Attribute Protocol Application error codes. Table 2.1: lists the set of error codes used in this service.

Name	Error Code	Description
Time Source Quality Too Low	0x80	Application error code
Incorrect Time Format	0x81	Application error code
Out of Range	0xFF	Common profile error code defined in CSS [2]

Table 2.1: Attribute Protocol error codes used by this service

2.5 GATT sub-procedure requirements

Requirements in this section represent a minimum set of requirements for a server. Other GATT sub-procedures may be used if supported by both the client and the server.

Table 2.2: summarizes additional GATT sub-procedure requirements beyond those required by all GATT servers.

GATT sub-procedure	Requirements
Characteristic Value Indication	M
Read Characteristic Descriptors	M
Write Characteristic Descriptors	M
Write Characteristic Value	C.1

Table 2.2: GATT sub-procedure requirements

C.1: Mandatory if the Current Elapsed Time characteristic can be written.



2.6 Declaration

The service universally unique identifier (UUID) shall be set to «Elapsed Time Service» as defined in [4].

ETS should be instantiated as a «Primary Service».

ETS should be instantiated only once in a device.

2.7 Behavior

ETS contains one characteristic that can be used to keep track of the current time of a server. The client can read the Current Elapsed Time characteristic to obtain the server's current time. Indications of this characteristic are used by the server to inform clients of changes to the current time of the server.

A client can also write to the Current Elapsed Time characteristic to set the time on the server, when supported by the server.

3 Service characteristics

This section defines the characteristic and descriptor requirements. Where a characteristic can be indicated or notified, a Client Characteristic Configuration descriptor must be included in that characteristic as required by [\[1\]](#).

Characteristic Name	Requirement	Mandatory Properties	Optional Properties	Security Permissions
Current Elapsed Time (see Section 3.1)	M	Read, Indicate	Write	none

Table 3.1: Characteristics for use with this service

3.1 Current Elapsed Time

The Current Elapsed Time characteristic informs clients of the current time and the capabilities of an Elapsed Time Server's clock. A client can update the current time of the server by writing to the Current Elapsed Time characteristic.

3.1.1 Characteristic format

The format of the Current Elapsed Time characteristic is shown in [Table 3.2](#):

Field	Requirement	Data Type	Size (in octets)	Description
Elapsed Time	M	struct	9	The field contains the current time of the server (see Section 3.1.1.1).
Clock Status	C.1	struct	1	The status of the server's clock (see Section 3.1.1.2).
Clock Capabilities	C.1	struct	1	The server's clock capabilities (see Section 3.1.1.3).

Table 3.2: Current Elapsed Time characteristic

C.1: Mandatory when the characteristic is read or indicated by the server, otherwise Excluded when written by a client.

3.1.1.1 Elapsed Time

When the Current Elapsed Time characteristic is read, the Elapsed Time field reports the server's current time. When written, the Elapsed Time field contains the time to be set as the current time on the server. The format of the Elapsed Time field is defined in GSS [\[3\]](#). Some examples of Elapsed Time values are given in [Appendix A](#).

3.1.1.2 Clock Status

[Table 3.3](#): defines the Clock Status field.

Bit #	Clock Status flag	Description
0	Clock needs to be set	Indicates that the server wants the clock to be set. This does not require the Current Elapsed Time to be writable by the client. The clock can, for example, only be set via the user interface (UI) of the server or by a specific other client.
1-7	Clock Status flags - RFU	Reserved for Future Use

Table 3.3: Clock Status

3.1.1.3 Clock Capabilities

Table 3.4: defines the Clock Capabilities field.

Bit #	Clock Capabilities flag	Description
0	Clock applies DST rules	Clock autonomously updates DST offset
1	Clock manages TZ changes	Clock autonomously updates TZ offset
2-7	Clock Capabilities flags - RFU	Reserved for Future Use

Table 3.4: Clock Capabilities

3.1.2 Characteristic behavior

3.1.2.1 Reading time

A client can understand the server's time and clock capabilities by reading the Current Elapsed Time characteristic. The flags in the Elapsed Time field tell the client which type of time and which resolution the server supports. The following types of time are supported in various resolutions:

- Coordinated Universal Time (UTC) time, with or without TZ/DST offset
- Local time, with or without TZ/DST offset
- Tick counter

The type of time a server supports, the TZ/DST offset support, and the time resolution are static.

The starting point for the Elapsed Time field for UTC or local time is 2000-01-01 00:00:00 as defined in GSS. For a tick counter, the starting point is implementation-dependent.

When reading the Current Elapsed Time characteristic, the Time Stamp Is From The Current Timeline flag in the Elapsed Time field is set to a value of 1. When using the Elapsed Time value as a time stamp to record an event, the Time Stamp Is From The Current Timeline flag may be set to a value of 0 when there has been a discontinuity in the timeline of the clock since the time was recorded. Examples of discontinuities in the timeline include a reset of the clock by a battery fault or a user setting the clock from the user interface (UI) of a device. In such situations, there is no relation between the current time as reported by the clock and the recorded Time value.



3.1.2.2 Reporting time changes

When the ETS server reconnects to a bonded client and when the current time has changed other than by natural progression, the ETS server shall send an indication of the Current Elapsed Time characteristic to the bonded client to facilitate a proper understanding of the server's clock.

When the ETS server is connected to a client and the current time changes other than by natural progression, the ETS server shall send an indication of the Current Elapsed Time characteristic to the client. When the server changes its time after a write by a client to the Current Elapsed Time characteristic, the server shall send an indication of the Current Elapsed Time characteristic to other connected clients. It is an implementation choice to send an indication to the client that performed the write or not.

The ETS server only sends indications of the Current Elapsed Time characteristic to a client when the client has enabled the Current Elapsed Time characteristic for indications.

3.1.2.3 Setting the time

If the Current Elapsed Time characteristic is writable, a client can set the time by writing to the Current Elapsed Time characteristic.

The server shall respond with an Attribute Protocol (ATT) Application Error code of Incorrect Time Format when the flags in the Flags field in the Elapsed Time field do not match the features as supported by the server as described in Section 3.1.2.1.

The server shall ignore the value of the Time Stamp Is From The Current Timeline flag when the time is set by a client. See GSS [3] for the definition of the flags of the Current Elapsed Time characteristic.

When writing the time, the server may respond with an Attribute Protocol Application Error under specific conditions as defined in Table 3.5: .

Error Code Name	Error Code	Condition
Time Source Quality Too Low	0x80	The time sync source used when setting the clock is of too low quality as judged by the server. For example, the server may consider a mobile network time source as lower quality than a Global Positioning System (GPS) time source. The conditions under which this error code is sent are implementation-specific.
Incorrect Time Format	0x81	The flags set in the value written to the Current Elapsed Time characteristic do not match what the server supports.
Out of Range	0xFF	The server judges the value as being out of range (e.g., when the value is from before the server production date). The conditions under which this error code is sent are implementation-specific.

Table 3.5: Time Setting Errors

4 SDP interoperability

If this service is exposed over BR/EDR, then it shall have the following SDP record:

Item	Definition	Type	Value	Status
Service Class ID List	-	-	-	M
Service Class #0	-	UUID	«Elapsed Time Service»	M
Protocol Descriptor List	-	Data Element Sequence	-	M
Protocol #0	-	UUID	«L2CAP»	M
Parameter #0 for Protocol #0	PSM	uint16	PSM = «ATT»	M
Protocol #1	-	UUID	«ATT»	M
BrowseGroupList	-	-	«PublicBrowseRoot»*	M

Table 4.1: SDP Record

* «PublicBrowseRoot» shall be present; however, other browse UUIDs may also be included in the list.

5 Acronyms and abbreviations

Acronym/Abbreviation	Meaning
ATT	Attribute Protocol
BR/EDR	Basic Rate/Enhanced Data Rate
CSS	Core Specification Supplement
DST	Daylight Saving Time
ETS	Elapsed Time Service
GATT	Generic Attribute Profile
GPS	Global Positioning System
GSS	GATT Specification Supplement
L2CAP	Logical Link Control and Adaptation Protocol
LSO	least significant octet
MSO	most significant octet
PSM	Protocol/Service Multiplexer
RFU	Reserved for Future Use
TZ	Time Zone
UI	user interface
UTC	Coordinated Universal Time
UUID	universally unique identifier

Table 5.1: Acronyms and abbreviations

6 References

- [1] Bluetooth Core Specification, Version 4.2 or later
- [2] Core Specification Supplement (CSS), v10 or later
- [3] GATT Specification Supplement (GSS), v10 or later
- [4] Bluetooth Assigned Numbers, <https://www.bluetooth.com/specifications/assigned-numbers/>



Appendix A Elapsed Time examples

The sections below give a few examples of Elapsed Time characteristic values.

A.1 Example 1: UTC time without TZ/DST offset in seconds

Field	Data Type	Size (in octets)	Example	Value (hex)	Little endian – transmission order
Flags	struct	1	Bit 0: Time is a tick counter: 0 Bit 1: Time is UTC: 1 Bit 2-3: Time resolution: 1 second - 00 Bit 4: TZ/DST offset is used: 0 Bit 5: Time stamp is from the current timeline: 1 Bit 6-7: RFU - 00 0b0010 0010	22	22
Time Value	uint48	6	2021-11-20 11:50:10	00 00 29 2B 9D 72	72 9D 2B 29 00 00
Time Sync Source Type	enumeration	1	Cellular network	06	06
TZ/DST Offset	int8	1	Not used	00	00

Table A.1: UTC time without TZ/DST offset in seconds

A.2 Example 2: UTC time with TZ/DST offset in milliseconds

Field	Data Type	Size (in octets)	Example	Value (hex)	Little endian – transmission order
Flags	struct	1	Bit 0: Time is a tick counter: 0 Bit 1: Time is UTC: 1 Bit 2-3: Time resolution: 1 millisecond - 10 Bit 4: TZ/DST offset is used: 1 Bit 5: Time stamp is from the current timeline: 1 Bit 6-7: RFU - 00 0b0011 1010	3A	3A
Time Value	uint48	6	2021-11-22 11:56:00.567 690897360567	00 A0 DC B1 16 B7	B7 16 B1 DC A0 00
Time Sync Source Type	enumeration	1	GPS	02	02
TZ/DST Offset	int8	1	Not used	00	00

Table A.2: UTC time with TZ/DST offset in milliseconds



A.3 Example 3: Local time

Field	Data Type	Size (in octets)	Example	Value (hex)	Little endian – transmission order
Flags	struct	1	Bit 0: Time is a tick counter: 0 Bit 1: Time is UTC: 0 Bit 2-3: Time resolution: 1 second - 00 Bit 4: TZ/DST offset is used: 0 Bit 5: Time stamp is from the current timeline: 1 Bit 6-7: RFU - 00 0b0010 0000	20	20
Time Value	uint48	6	2021-11-20 11:50:10	00 00 29 2B 9D 72	72 9D 2B 29 00 00
Time Sync Source Type	enumeration	1	Manual	04	04
TZ/DST Offset	int8	1	Not used	00	00

Table A.3: Local time

A.4 Example 4: Tick counter

Field	Data Type	Size (in octets)	Example	Value (hex)	Little endian – transmission order
Flags	struct	1	Bit 0: Time is a tick counter: 1 Bit 1: Time is UTC: 0 Bit 2-3: Time resolution: 100 microseconds - 11 Bit 4: TZ/DST offset is used: 0 Bit 5: Time stamp is from the current timeline: 1 Bit 6-7: RFU - 00 0b0010 1101	2D	2D
Time Value	uint48	6	2 ³² 4294967296	00 01 00 00 00 00	00 00 00 00 01 00
Time Sync Source Type	enumeration	1	Other / not used	00	00
TZ/DST Offset	int8	1	Not used	00	00

Table A.4: Tick counter